

Leggett-Garg Inequality-based Quantum Computers Quality Benchmark

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Motivation

European Quantum Strategy

Our research are aligned with the European Quantum Strategy.

- The need for independent benchmarking tools.
- Testing under-explored properties of quantum devices.
- Unexplained error sources found in previous research.

Constellation-topology devices

With the advent of Constellation-topology devices, more attention should be given to devices with central resonators.

- Software for devices with central resonator (e.g. Qubit Selector)
- Benchmarks that promote high inter-qubit connectivity.
- Dedicated implementation of weak measurements using pulse-level access.

Executive summary

Primary research

We propose Leggett-Garg inequality-based benchmark that is:

- Highly scalable (different topologies, parallelization),
- Versatile: application to benchmarking partially entangling gates and pulse control,
- Extendable (pulse-controlled weak coupling, more qubits...),
- Robust to noise.

Technical results

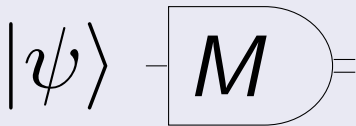
We present repository with the code we produced. Therein:

- Implementation of benchmark circuits, experimental pipelines and analysis scripts.
- Qubit Selector for Star-shaped topology devices. Expandable to Constellation-topology devices.
- Simulators based on the historical calibration data.

Projective vs weak measurement

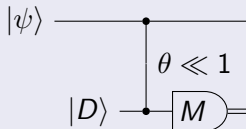
Projective (strong) measurement

- Direct
- Accurate (in theory)
- Disturbing (decoherence)



Weak measurement

- Indirect (weak coupling)
- Inaccurate (deliberately)
- Small decoherence



No free lunch

- Information gain $> \theta$
- Decoherence $> \theta^2$

The catch: Measurement strength θ calibration

$$a_{\text{raw}} = \theta a_{\text{physical}}$$

Application: Leggett-Garg inequality violation

Leggett-Garg (LG) Inequality

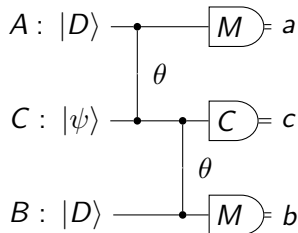
$$\langle a \rangle + \langle b \rangle - \langle ab \rangle \leq 1, |a| \leq 1, |b| \leq 1$$

LG-based benchmark inequality

$$Q = \frac{\langle a|c \rangle \langle b|c \rangle}{\langle ab|c \rangle} \leq 1,$$

where $\langle ab|c \rangle = \langle abc \rangle / p(c = 1)$.

- measurement strength-agnostic
- postselection $c = 0, 1$,
- $A \equiv B$ (identical, independent)



Detectors identity and independence ($A \equiv B$)

Partially verifiable by $D = 0$, where $D := \langle a \rangle \langle bc \rangle - \langle b \rangle \langle ac \rangle$.

A test with strength dependence on $\lambda = \sin \theta$

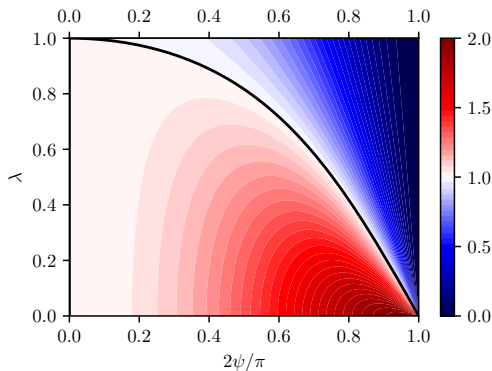
Setup

$$A = B = Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$C = |\psi_{-}\rangle\langle\psi_{-}|$$

$$|\psi\rangle = |\psi_{+}\rangle$$

$$|\psi_{\pm}\rangle = \cos \frac{\psi}{2} |1\rangle \pm \sin \frac{\psi}{2} |0\rangle$$



Benchmark scores

- Simple: $B_S = (Q - 1)/\sigma = n_\sigma$
- Normalized (one-tailed confidence): $B_N = \frac{1}{2} \text{erf} \left(\frac{n_\sigma}{\sqrt{2}} \right)$

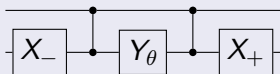
Weak coupling implementation

Outline

- Two qubit gate $U \sim I$
- Basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

Weak coupling with IQM-native gates

We use two CZ gates sandwiching Y_θ gate.



Notation

$$X = |0\rangle\langle 1| + |1\rangle\langle 0|$$

$$Y = i|1\rangle\langle 0| - i|0\rangle\langle 1|$$

$$V_\theta = \exp(-i\theta V)$$

$$V_\pm = V_{\pm\pi/2}$$

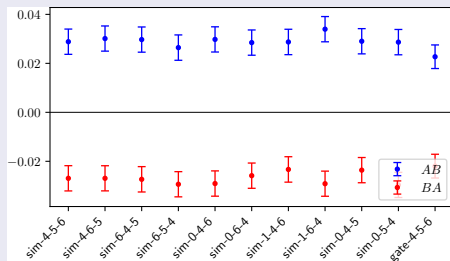
Alternative: Fractional gates

Weak coupling implementation is native on IBM / IonQ devices with the fractional RZZ gate.

$$RZZ(\theta) = \text{diag}(1, e^{i\theta}, e^{i\theta}, 1)$$

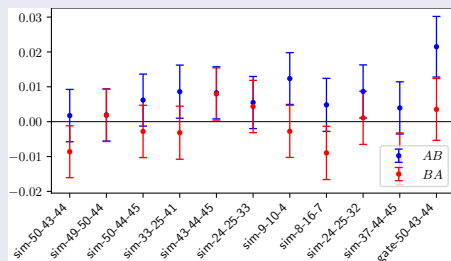
Experiments: Detectors independence ($\theta = 0.1$)

IQM Sirius



- CAB qubits specified
- 10^6 shots per qubits set
- D within $> 5 \sigma$ from 0
- Simulator close to real device

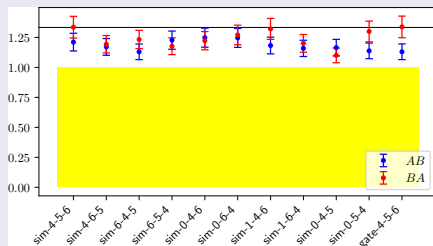
IQM Emerald



- CAB qubits specified
- 10^6 shots per qubits set
- D within $< 1 \sigma$ from 0
- Simulation far from real device

Experiments: Benchmark score ($\theta = 0.1$)

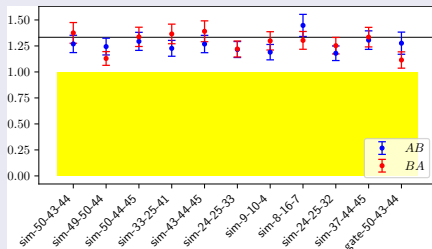
IQM Sirius



Benchmark values (on device):

- $B_S^{AB} = 1.9954$, $B_S^{BA} = 3.7494$
- $B_N^{AB} = 0.9770$, $B_N^{BA} = 0.9999$

IQM Emerald



Benchmark values (on device):

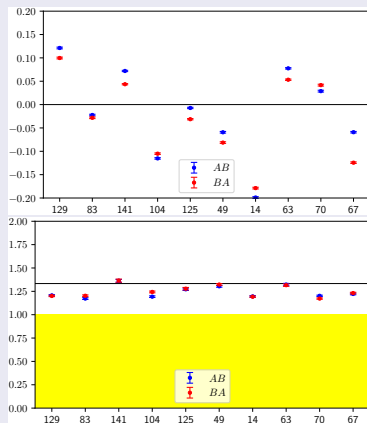
- $B_S^{AB} = 2.5935$, $B_S^{BA} = 1.4625$
- $B_N^{AB} = 0.9253$, $B_N^{BA} = 0.9282$

Shots number

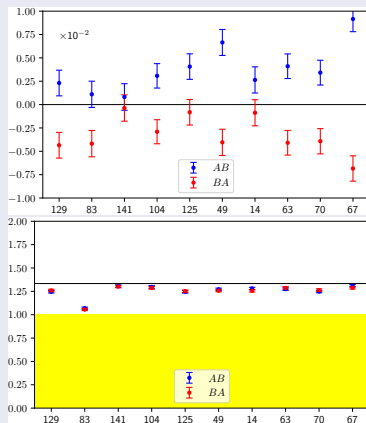
The results could be improved with better statistics.

Experiments: Different providers ($\theta = 0.1$)

IBM Kingston (RZZ, $2 \cdot 10^7$ shots)



IBM Kingston (CZ, $2 \cdot 10^7$ shots)



IonQ Forte

IonQ Forte 1: 1.272 ± 0.035 ($A \rightarrow B$), 1.289 ± 0.035 ($B \rightarrow A$)

Failed Experiment: IQM Pulla

- IQM Sirius
- # shots = 10^6
- qubits: [4, 5, 6] (best set)
- Measurement pulse amplitude halved for all qubits

Conclusions

- Naive pulse modification lead to poor results
- In some cases, benchmark inequality may still be violated

Detectors validation

- $D_{AB} = -0.000871 \pm 0.005872$
- $D_{BA} = -0.000068 \pm 0.005807$

Benchmark Scores

- $Q_{AB} = 0.8012 \pm 0.3660$
- $Q_{BA} = 1.8411 \pm 1.9617$
- $B_S^{AB} = -0.543063$
- $B_N^{AB} = 0.293543$
- $B_S^{BA} = 0.428756$
- $B_N^{BA} = 0.665950$

Impact and directions

Impact

- New benchmarking protocol for different properties of the device
- Benchmarking partially entangling gates and pulse control
- Robustness to noise – no error correction needed
- Minimal disturbance: $\theta = 0.1 \rightarrow$ disturbance $1/400$
- Device-agnostic (highly scalable)

Research directions

- Test benchmark limits with sufficient statistics ($\sigma \sim 1/\theta^2\sqrt{N}$)
- Weak coupling implementation using pulse-level control e.g. creating continuous families of entangling gates ($CZ \rightarrow \text{diag}(1, 1, 1, e^{\pm i\theta})$)
- Avoid postselection (full noninvasiveness)
- Scaling up the test. Extend qubits sets on star topology devices. Benchmark score extension for whole device.
- Investigate the reason of D increase in RZZ protocol version.

Thank you for your attention!

Special thanks to:

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 - Jakub Tworzydło
 - Josep Batle
- Device providers: IQM, IBM, IonQ
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Repository



Our LG manuscripts

